

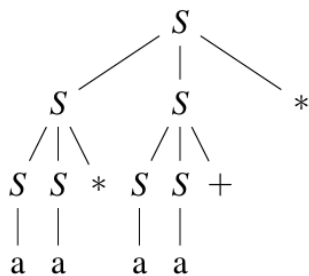
Assignment 3

Name:

1 Required Exercises

Exercise 1

1. **No, it isn't.** (Either a single 'a' alone, or ending with '+' or ending with '*'.)
2. $S \Rightarrow_{lm} SS \Rightarrow_{lm} SS * S \Rightarrow_{lm} aS * S \Rightarrow_{lm} aa * S \Rightarrow_{lm} aa * SS \Rightarrow_{lm} aa * aS \Rightarrow_{lm} aa * aa + *$
3. $S \Rightarrow_{rm} SS \Rightarrow_{rm} SSS \Rightarrow_{rm} SSa \Rightarrow_{rm} Saa \Rightarrow_{rm} SS * aa \Rightarrow_{rm} Sa * aa \Rightarrow_{rm} aa * aa + *$
- 4.



- $$5. S \rightarrow aS', S' \rightarrow S + S' | S * S' | \varepsilon.$$

Exercise 2

1. (a) Calculate the FIRST and FOLLOW sets for the grammar.

$$FIRST(S) = \{a\}$$

$$FOLLOW(S) = \{+, \$\}$$

$$FIRST(B) = \{a, \varepsilon\}$$

$$FOLLOW(B) = \{+, \$\}$$

$$FIRST(aB) = \{a\}$$

$$FIRST(S + B) = \{a\}$$

$$FIRST(\varepsilon) = \{\varepsilon\}$$

- (b) Construct the predictive parsing table for the grammar.

NON-TERMINAL	INPUT SYMBOL		
	a	+	\$
S	$S \rightarrow aB$		
B	$B \rightarrow S + B$	$B \rightarrow \varepsilon$	$B \rightarrow \varepsilon$

- 2. Yes, it is.**

$$FIRST(S + B) \cap FIRST(\varepsilon) = \emptyset$$

$$FIRST(S + B) \cap FOLLOW(B) = \emptyset$$

3. Yes, it can.

MATCHED	STACK	INPUT	ACTION
	$S\$$	$aaaa+++\$$	
	$aB\$$	$aaaa+++\$$	$output\ S \rightarrow aB$
a	$B\$$	$aaa+++\$$	$match\ a$
a	$S+B\$$	$aaa+++\$$	$output\ B \rightarrow S+B$
a	$aB+B\$$	$aaa+++\$$	$output\ S \rightarrow aB$
aa	$B+B\$$	$aa+++\$$	$match\ a$
aa	$S+B+B\$$	$aa+++\$$	$output\ B \rightarrow S+B$
aa	$aB+B+B\$$	$aa+++\$$	$output\ S \rightarrow aB$
aaa	$B+B+B\$$	$a+++\$$	$match\ a$
aaa	$S+B+B+B\$$	$a+++\$$	$output\ B \rightarrow S+B$
aaa	$aB+B+B+B\$$	$a+++\$$	$output\ S \rightarrow aB$
$aaaa$	$B+B+B+B\$$	$+++\$$	$match\ a$
$aaaa$	$+B+B+B\$$	$+++\$$	$output\ B \rightarrow \epsilon$
$aaaa+$	$B+B+B\$$	$++\$$	$match\ +$
$aaaa+$	$+B+B\$$	$++\$$	$output\ B \rightarrow \epsilon$
$aaaa++$	$B+B\$$	$+\$$	$match\ +$
$aaaa++$	$+B\$$	$+\$$	$output\ B \rightarrow \epsilon$
$aaaa+++$	$B\$$	$\$$	$match\ +$
$aaaa+++$	$\$$	$\$$	$output\ B \rightarrow \epsilon$

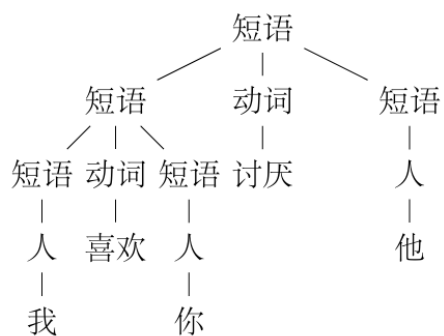
Accept. The top of stack is \$.

2 Optional Exercises

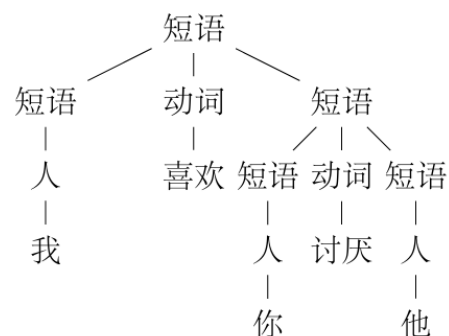
Exercise 1

Yes, it is ambiguous.

e.g. 我喜欢你讨厌他



(a) Parse tree 1



(b) Parse tree 2

Figure: Two parse trees for the ambiguous sentence